Travel Document Analyzer – Backend Development Plan

Goal:

To develop a backend system that analyzes user-submitted travel documents (e.g. passport, visa) using OCR and verifies them against country-specific checklists to flag missing or invalid documents.

Week 1 – Backend Foundations

Build core REST API structure using Flask
- Set up project structure and environment
- Implement endpoints for retrieving required documents and validating user submissions
- Use JSON for initial rule storage
- Test with Postman or mock frontend

Tech: Flask, Python dictionaries, JSON

Week 2 – OCR Integration

Accept image uploads and extract raw text
- Integrate pytesseract and Tesseract OCR engine
- Add /extract-text endpoint to process image files
- Organize uploaded files in local storage

Tech: Tesseract, pytesseract, PIL, Flask uploads

Week 3 – Text Parsing & Field Extraction

Extract structured fields from OCR text output
- Use regex and string logic to identify passport numbers, visa data, expiry dates
- Normalize different formats into a standard format

Tech: Python regex, string handling, datetime

Week 4 – Validation & Document Matching Logic

Compare extracted fields against expected checklist
- Match OCR results with country-specific document rules
- Return detailed reports on missing/invalid items

Tech: Python sets, logic, date parsing

Week 5 – Rule Management & Optional DB Integration

Replace JSON with a dynamic rule database (optional)
- Use SQLite to manage checklists based on destination, purpose, nationality
- Create admin function to update rule data

Tech: SQLite3, SQL basics

Week 6 – Stability, Security & Error Handling

Harden backend and improve robustness
- Add validation and exception handling
- Handle missing files, bad input, and edge cases
- Clean temp files and sanitize uploads

Tech: Flask error handling, file validation

Week 7 – Documentation & Final Testing

Prepare system for final presentation and handoff
- Write complete README and API usage documentation
- Final testing with sample data and frontend
- Prepare a short demo of the working backend

Deliverables: working backend, test cases, documentation

Tech Stack Summary

Layer	Tools
API Framework	Flask
OCR Engine	Tesseract via pytesseract
File Handling	Pillow, Flask upload
Data Rules	JSON → SQLite (optional)
Testing	Postman, Python requests
Parsing	Regex, datetime, string logic